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MATH 315

Question 4:

(a). From the previous question and the ODE that we got, let's assume that the mass and damping constant is set to 1 and the length is equal to gravity.

The equilibriums are at $(0,0)$ and $(\pi,0)$.

(b). The linear system for the first equilibrium,

$A = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$ which will give us the eigenvectors of $\frac{1}{2} \pm \sqrt{3}i$.

$\begin{bmatrix} -1 & -1 \end{bmatrix}$

For the second case, we get $A = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$

$\begin{bmatrix} 1 & -1 \end{bmatrix}$ which would give us the eigenvectors $\frac{-1 \pm \sqrt{5}}{2}$.

With these linear systems, the spirals on the phase portrait approach the equilibriums toward the origin of the graph.